Research Proposal - Applied Deep Learning

Few-Shot Adaptation of Cheff: Chest X-ray Generation for Specialized Medical Domains

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Abstract

Diffusion models have emerged as the predominant tool to create synthetic medical image data. Among them Cheff, a latent diffusion model, has demonstrated state-of-the-art capability in generating highly realistic, 1-megapixel chest X-ray (CXR) images. However, the successful deployment of downstream clinical AI applications often demands highly adaptable synthetic images to adapt to shifts in patient demographics, imaging protocols, or pathology prevalence. This is why we propose a framework for Parameter-Efficient Fine-Tuning (PEFT), designed to efficiently adapt Cheff's generalized representations to these unseen domains. The resulting PEFT-adapted Cheff framework will serve as a practical tool for clinical researchers, enabling the creation of synthetic data to power more specialized and robust AI assistance systems in medicine.

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Motivation

The development of clinical AI assistance tools is constrained by data quality and availability problems such as rare pathologies, noisy labels and inherent demographic biases. Diffusion models have emerged as an impactful tool to overcome these issues by creating synthetic medical images. The Cheff model as introduced by Weber et al. (2023), a large-scale cascaded latent diffusion architecture, has demonstrated the ability to synthesize highly realistic, 1-megapixel chest X-ray (CXR) images. Trained on the extensive MaCheX collection, Cheff supports both unconditional and text-conditioned generation via radiological reports, allowing fine-grained control over the synthesis process. However, for the training and deployment of many clinical AI applications, synthetic images from highly specialized or scarce domains are necessary. The following domains are identified as initial use cases for adaptation:

Rare or Unseen Pathologies - few-shot fine-tuning enables extrapolation of limited disease cases to create sufficient training data.
Imaging Protocols - adaptation to site-specific scanner characteristics improves compatibility with clinical infrastructure.
Demographic Adaptation - addressing population imbalances in publicly available datasets enables fairer and more robust AI systems.
Patient Anatomy Personalization - while generative models should not serve diagnostic purposes, they can support counterfactual visualization for individual patients.

To enable the generation of diverse and clinically meaningful synthetic datasets across various scarce or specialized domains, we propose a Parameter-Efficient Fine-Tuning (PEFT) framework to adapt Cheff's foundational representations in a resource-efficient manner. To this end, we will evaluate the general feasibility of adaptation in the identified domains, quantify the corresponding adaptation effort required for each case and identify the architectural components most critical for effective adaptation.

Related Work

Research on medical image synthesis has progressively shifted to diffusion-based models. Ali et al. (2023) and Chambon et al. (2022) investigate how large pretrained diffusion foundation models such as DALLE-2 and Stable Diffusion can be fine-tuned for the medical imaging domain to evaluate their ability to generate realistic and clinically meaningful images, but with limited success and high computational cost.
Blüthgen et al. (2025) emphasize that medical imaging data and their textual modalities differ fundamentally from natural images and captions, concluding that multimodal models trained solely on natural image–text pairs fail to generalize well to the medical domain and therefore require a substantial domain shift to be applied successfully in medical contexts. With the introduction of the MaCheX dataset collection and the development of Cheff Weber et al. (2023), a domain-specific, large-scale diffusion model for medical imaging has become available. Such domain-specific foundation models can serve as a more efficient and suitable base for adaptation to smaller medical subdomains, such as specific pathologies or demographic groups, which are required in clinical applications.

Yet, no prior work has systematically examined the few-shot learning capabilities of such models for generating synthetic training data in these specialized contexts. This research addresses that gap by exploring the efficient domain adaptation of Cheff to closely related medical imaging subdomains. Leveraging its smaller model size and domain-specific pre-training, we aim to investigate whether Cheff enables computationally and data-efficient fine-tuning for high-quality, domain-relevant synthetic image generation.

Methodology

The publicly available Cheff diffusion model will serve as our pretrained foundation model. Our primary objective is to apply Low-Rank Adaptation (LoRA) fine-tuning as introduced by Hu et al. (2022) to efficiently adapt Cheff to specific medical subdomains. We anticipate that it will be necessary to adapt the number of training examples and other learning parameters dynamically, depending on the characteristics and distributional shift of each target domain. By introducing domain-specific tokens in the text conditioning of the semantic generator, the model is expected to learn associations between these tokens and their corresponding target domains, enabling it to reproduce domain-specific characteristics when prompted.

Our initial hypothesis is that LoRA adapters should primarily target the semantic generator component of Cheff, as well as cross-attention layers responsible for text and report conditioning. These components capture the general semantic representations that are subsequently refined to high resolution by the super-resolution module. We further aim to quantitatively assess the quality of the synthesized images using distance-based metrics such as the Fréchet Inception Distance (FID) or Kernel Inception Distance (KID).

Details of Implementation

We plan to work on the project at a steady pace throughout the semester, with a final submission targeted for March 2026 in the form of a research paper. The project can be structured into five main phases.

Phase 1: Assess the required computational resources, select suitable domain-specific datasets, and determine the necessary preprocessing steps to ensure compatibility with Cheff's input requirements.

Phase 2: Load and set up the pretrained Cheff model, followed by the integration of LoRA adapters into the model architecture.

Phase 3: Conduct Parameter-Efficient Fine-Tuning (PEFT) across multiple domains, quantifying the minimal adaptation effort required for effective domain transfer.

Phase 4: Evaluate the fidelity and quality of the generated synthetic images using quantitative metrics such as the Fréchet Inception Distance (FID) and Kernel Inception Distance (KID) to validate the efficacy of the proposed fine-tuning scheme.

Phase 5: Formulate the seminar paper and clean the code repository.

References

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